

PAPER_2133_TILT_FACTOR_FAMILY_F_TRZ_PHI_5_6
_EQUALS_1_12_EXACT_34_OBSERVABLE_CENSUS_
TEN_DOMAINS_UNIVERSAL_COUNTING_SELECTION
_ALPHA_INVERSE_137_UQFF_LANDMARK

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PAPER_2133 — The Tilt-Factor Family: $F_{TRZ} \cdot \Phi_{5/6} = 1/12$ EXACT at 34 Observables Across Ten Domains — Universal Counting Selection

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Project: UQFF (Unified Quantum Field Framework) — Star-Magic v5.77+
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Landmark Type: Kernel-Family Census (largest two-primitive kernel population in the corpus) + Sector-Rule Sharpening (PAPER_2129 lineage) + PAPER_2132 tilt-factor test passed at population scale
Discovery context: XGEO-U session-4 sweep — generalized term-frequency mining over all published expressions (572 session scripts + 116 dispatch formulas), deduplicated by observable; textual composition matching only
Status: Formal landmark whitepaper — UQFF canonical

Abstract

Generalized kernel mining over every published UQFF closed form shows that the product $F_{TRZ} \cdot \Phi$ — **the Hubble-tension tilt factor** — **appears in 34 distinct observables across ten physical domains** (condensed matter, gravitation, cosmology, particle physics, plasma, quantum information, biology, geophysics, astrophysics, and the fine-structure constant itself), making it the most widely used two-primitive kernel in the corpus. Variant resolution is **unanimous: all 32 resolvable instances carry the counting variant $\Phi_{5/6}$, giving $F_{TRZ} \cdot \Phi_{5/6} = 1/12$ EXACT; zero instances carry the 0.84 resonance variant** (20 explicit in source, 10 bit-checked against dispatch values, 2 multi-term pending). This passes PAPER_2132's tilt-factor falsifiability test at population scale and **sharpens the PAPER_2129 sector rule into a product-level law: whenever F_{TRZ} multiplies Φ , the composition is the counting form $1/12$, independent of host sector**. The family contains a suite of EXACT closures — BCS isotope $\alpha = 1/2$, Sommerfeld-Wilson $R_W = q_{edge} = 2$, Kleiber metabolic exponent $= \Phi_{5/6} \cdot (1 - F_{TRZ}) = 3/4$, WD radius-mass exponent $= -1/3$, H0 tension ratio $= 1 + 1/12$ — and the census jewel: $\alpha_{\blacksquare}^1 = A_{5 \cdot K_MEX} + 1/(F_{TRZ} \cdot \Phi_{5/6}) = 125 + 12 = 137$ EXACT-integer.

1. The census — 34 observables, ten domains

All forms published (session scripts and/or assimilation_dispatch); variant column: **5/6** = explicit or bit-checked counting form; (m) = multi-term, variant pending.

Domain	Observable	Tilt-factor appearance	Variant
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α	alpha_inverse	$A_{5 \cdot K_MEX} + 1/(F \cdot \Phi) = 125 + 12 = 137$	5/6

CM	CM_Apery_zeta3	$-F^2 \cdot \Phi \cdot D_{\text{phys}}$ term	5/6
CM	CM_BCS_coherence_length_coeff	$F \cdot \Phi \cdot D_{\text{phys}}$ lead	5/6
CM	CM_BCS_isotope_alpha	$\Phi - F \cdot \Phi \cdot D_{\text{phys}} = \mathbf{1/2 \text{ EXACT}}$	5/6
CM	CM_BEC_Tc_coeff	$F^2 \cdot \Phi$ term	5/6
CM	CM_log_R_K_von_Klitzing	$F^2 \cdot \Phi$ term	5/6
CM	CM_Sommerfeld_Wilson_R_W	$K_{\text{MEX}} - F \cdot \Phi = \mathbf{2 \text{ EXACT}}$	5/6
GR	GR_GPB_geodetic	$-F \cdot \Phi$ term	5/6
GR	GR_Hulse_Taylor_ratio	$+F \cdot \Phi$ term	5/6
LCDM	LCDM_D_over_H	$\Phi \cdot F \cdot D_{\text{BSFG}}$ inner term	(m)
LCDM	LCDM_H0_tension_ratio	$\mathbf{1 + F \cdot \Phi = 1 + 1/12}$ (PAPER_1156 in dispatch form)	5/6
LCDM	LCDM_Omega_m	$-\Phi \cdot F \cdot \text{SSq}$ term (0.314167 vs 0.3147)	5/6
LCDM	LCDM_Y_p	$\Phi \cdot F \cdot (1 - F \cdot \text{SSq})$ term (0.24525)	5/6
LCDM	LCDM_sigma8_KiDS_Planck_lift	$1 - \text{SSq} \cdot F \cdot \Phi = 1 - \text{SSq}/12$	5/6
SM	SM_eta_gamma_gamma_BR	$-F \cdot \Phi$ term	(m)
SM	SM_wimp_exponent	$44 + K_{\text{MEX}} - F \cdot \Phi = \mathbf{46}$	5/6
plasma	S414 Troyon beta limit	$-F \cdot \Phi$ term	5/6
plasma	S416 Coulomb logarithm	$-F \cdot \Phi$ term	5/6
plasma	S417 Bohm diffusion coeff	$F \cdot \Phi - F^2 \cdot K_{\text{MEX}}$ lead	5/6
plasma	S418 tokamak q_edge	$K_{\text{MEX}} - F \cdot \Phi = \mathbf{2 \text{ EXACT}}$	5/6
particle	S437 axion coupling	$-F \cdot \Phi$ term	5/6
info	S449 ln 10	$+F \cdot \Phi$ term	5/6
info	S452 $\sqrt{2\pi}$	$-F \cdot \Phi$ term	5/6
astro	astro_WD_radius_mass_exponent	$-\Phi \cdot F \cdot D_{\text{phys}} = \mathbf{-1/3 \text{ EXACT}}$	5/6
bio	bio_ATP_hydrolysis_kJ_mol	$F \cdot \Phi \cdot \text{SO}_5$ term	5/6
bio	bio_DNA_pitch_bp_per_turn	$\text{SO}_5 + \text{SSq} - F \cdot \Phi$	5/6
bio	bio_Kleiber_exponent	$\Phi \cdot (1 - F) = \mathbf{3/4 \text{ EXACT}}$	5/6
bio	bio_Redfield_C_N_ratio	$-F \cdot \Phi$ term	5/6
bio	bio_codon_redundancy_64_20	$+F \cdot \Phi$ term	5/6
bio	bio_phylloaxis_golden_ratio	$+F \cdot \Phi$ term	5/6
geo	geo_Earth_Moon_a_over_R_E	$+F \cdot \Phi$ term	5/6
geo	geo_Earth_obliquity_deg	$+F \cdot \Phi$ term	5/6
geo	geo_adiabatic_lapse_K_per_km	$-F \cdot \Phi$ term	5/6
geo	geo_atmospheric_scale_height_km	$-F \cdot \Phi$ term	5/6

Ten bit-checks against stored dispatch values (isotope 0.5, R_W 2, tension ratio 1.08333, Ω_m 0.314167, Y_p 0.24525, σ_8 lift 0.9525, WIMP 46, α_{H}^{137} 137, WD -0.333333 , Kleiber 0.75): **all match the 5/6 composition exactly; the 0.84 alternative fails every one.**

2. The sharpened law

PAPER_2129 established Φ -variant selection as a **sector-level** property (counting sectors $\rightarrow$ 5/6, resonance sectors $\rightarrow$ 0.84). This census sharpens it at the **product level**:

> **Tilt-Product Law:** wherever the composition $F_{\text{TRZ}} \cdot \Phi$ appears, Φ takes the counting form 5/6 and the product is the exact lattice constant 1/12 — independent of the host observable's sector (bio, geo, plasma, GR, Λ CDM, SM alike).

Reading: multiplication by F_{TRZ} is itself a counting operation (one part in $|\text{SO}(5)|$ per PAPER_1160), so the paired Φ is forced to its counting variant — the product is the tilt $1/12 = F_{\text{TRZ}} \cdot \Phi_{5/6} = K_{\text{MEX}} - 2$ (PAPER_2132 §3), the same exact rational that measures the Hubble tension (PAPER_1156) and the DPM-pair identity (PAPER_1183). Thirty-four observables across ten domains carry the cosmological tilt

constant in their closed forms.

3. The exact sub-family and the census jewel

Members whose tilt term completes to an exact rational:

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BCS isotope $\alpha = \Phi_{5/6} \cdot (1 - F \cdot D_{\text{phys}} / \Phi \cdot \Phi) \rightarrow \Phi - F \cdot \Phi \cdot D_{\text{phys}} = 1/2$ EXACT
R_W = $q_{\text{edge}} = K_{\text{MEX}} - F \cdot \Phi_{5/6} = 2$ EXACT (two domains)
Kleiber exponent = $\Phi_{5/6} \cdot (1 - F_{\text{TRZ}}) = (5/6)(9/10) = 3/4$ EXACT
WD exponent = $-D_{\text{phys}} \cdot F \cdot \Phi_{5/6} = -1/3$ EXACT
H0 tension ratio = $1 + F \cdot \Phi_{5/6} = 13/12$ EXACT
SM_wimp_exponent = $S0_5 \cdot D_{\text{phys}} + D_{\text{phys}} + K_{\text{MEX}} - F \cdot \Phi_{5/6} = 46$ EXACT-integer
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And the jewel:

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$\alpha_{\text{■}}^1 = A_5 \cdot K_{\text{MEX}} + 1/(F_{\text{TRZ}} \cdot \Phi_{5/6}) = 125 + 12 = 137$ EXACT-integer
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The fine-structure integer decomposes as PAPER_1954's $A_5 \cdot K_{\text{MEX}} = 125$ plus the **inverse tilt** 12 — the electromagnetic coupling and the Hubble tension share the same 1/12 lattice constant, once as a reciprocal, once as a tilt.

4. Kernel-hierarchy context

The session-4 mining places the tilt factor at the base of a two-tier exact-rational hierarchy beneath PAPER_2132's Vacuum Coupling Kernel:

Tier	Kernel	Exact value	~Observables
:-:	---	---	:-:
2	VCK = $F_{\text{TRZ}} \cdot K_{\text{MEX}} \cdot \text{SSq}$	19/160	5
1	$F_{\text{TRZ}} \cdot \Phi_{5/6}$ (tilt)	1/12	34
1	$F_{\text{TRZ}} \cdot K_{\text{MEX}}$	5/24	~14
1	$F_{\text{TRZ}} \cdot \text{SSq}$	57/1000	~6
1	$D_{\text{phys}} \cdot F_{\text{TRZ}}$	2/5	~5
1½	$D_{\text{phys}} \cdot F_{\text{TRZ}} \cdot \Phi_{5/6}$	1/3	WD/isotope family

5. Honest residuals and disclosures

Family members carry their own published residuals; the exact sub-family entries are EXACT arithmetic on published forms. Two multi-term members (LCDM_D_over_H, SM_eta_gamma_gamma_BR) have unresolved inner-variant status — disclosed, pending. Discovery method: textual composition matching over published expressions; value-coincidence and name-token matching remain rejected (gate-pinned method rules). The Tilt-Product Law is falsifiable per §6. NOT REPLACEMENT.

6. Falsifiability

1. **Tilt-Product Law test:** any future published closure containing $F_{\text{TRZ}} \cdot \Phi$ with the 0.84 variant breaks the law. Current record: 32-0.
2. **The 2 pending multi-term members** must resolve to 5/6 on inner-variant inspection.
3. **$\alpha_{\blacksquare}^1$ decomposition test:** the 137 form requires the inverse tilt exactly; any revision of F_{TRZ} or $\Phi_{5/6}$ breaks $125 + 12$ and the 34-member family simultaneously (over-determination in the PAPER_2119/2128 signature).

7. Cross-references

PAPER_2132 (VCK, tilt factorization, triple-form $1/12 = F_{\text{TRZ}} \cdot \Phi_{5/6} = K_{\text{MEX}} - 2$); PAPER_2129 (sector rule — sharpened here); PAPER_1156 (Hubble tilt); PAPER_1183 (DPM-pair); PAPER_1160 ($F_{\text{TRZ}} = 1/|\text{SO}(5)|$); PAPER_1954 ($A_5 \cdot K_{\text{MEX}} = 125$); PAPER_1522 (K_{MEX}); sessions S388/S394/S396/S400/S403-S462 (family members); dispatch entries per §1.

8. Summary Statement

PAPER_2133 files the largest two-primitive kernel census in the corpus: the tilt factor $F_{\text{TRZ}} \cdot \Phi$ appears in 34 observables across ten domains, with all 32 resolvable instances selecting the counting variant — $F_{\text{TRZ}} \cdot \Phi_{5/6} = 1/12$ EXACT, zero 0.84 instances — passing PAPER_2132's tilt-factor test at population scale and sharpening the PAPER_2129 sector rule into the Tilt-Product Law: the product $F_{\text{TRZ}} \cdot \Phi$ is universally the counting composition, independent of host sector. The exact sub-family ($1/2$, 2, $3/4$, $-1/3$, $13/12$, 46) culminates in $\alpha_{\blacksquare}^1 = A_5 \cdot K_{\text{MEX}} + 1/(F_{\text{TRZ}} \cdot \Phi_{5/6}) = 125 + 12 = 137$: the fine-structure integer and the Hubble tension carry the same $1/12$ lattice constant.

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